rather than by the formal constraints of the sampling theorem. Aliasing *can* occur in images, but it is generally thought of as a nuisance rather than a major problem. For instance, pinstriped suits look terrible on television because the repetitive pattern is greater than the Nyquist frequency. The aliased frequencies appear as light and dark bands that move across the clothing as the person changes position.

A "typical" digital image is composed of about 500 rows by 500 columns. This is the image quality encountered in television, personnel computer applications, and general scientific research. Images with fewer pixels, say 250 by 250, are regarded as having unusually poor resolution. This is frequently the case with new imaging modalities; as the technology matures, more pixels are added. These low resolution images look noticeably unnatural, and the individual pixels can often be seen. On the other end, images with more than 1000 by 1000 pixels are considered exceptionally good. This is the quality of the best computer graphics, high-definition television, and 35 mm motion pictures. There are also applications needing even higher resolution, requiring several thousand pixels per side: digitized x-ray images, space photographs, and glossy advertisements in magazines.

The strongest motivation for using lower resolution images is that there are *fewer* pixels to handle. This is not trivial; one of the most difficult problems in image processing is managing massive amounts of data. For example, one second of digital audio requires about eight *kilobytes*. In comparison, one second of television requires about eight *Megabytes*. Transmitting a 500 by 500 pixel image over a 33.6 kbps modem requires nearly a minute! Jumping to an image size of 1000 by 1000 *quadruples* these problems.

It is common for 256 **gray levels** (quantization levels) to be used in image processing, corresponding to a single byte per pixel. There are several reasons for this. First, a single byte is convenient for data management, since this is how computers usually store data. Second, the large number of pixels in an image compensate to a certain degree for a limited number of quantization steps. For example, imagine a group of adjacent pixels alternating in value between digital numbers (DN) 145 and 146. The human eye perceives the region as a brightness of 145.5. In other words, images are very *dithered*. Third, and most important, a brightness step size of 1/256 (0.39%) is smaller than the eye can perceive. An image presented to a human observer will not be improved by using more than 256 levels.

However, some images need to be stored with more than 8 bits per pixel. Remember, most of the images encountered in DSP represent nonvisual parameters. The acquired image may be able to take advantage of more quantization levels to properly capture the subtle details of the signal. The point of this is, don't expect to human eye to see all the information contained in these finely spaced levels. We will consider ways around this problem during a later discussion of brightness and contrast.

The value of each pixel in the digital image represents a small *region* in the continuous image being digitized. For example, imagine that the Venus